

Counting On to Compare Two Sets

Answer Key

Kindergarten–Grade 1 Counting and cardinality

Quick Answers

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|-------|------|------|-------|
| 1. 8 | 4. 3 | 7. — | 10. — |
| 2. 14 | 5. — | 8. — | |
| 3. — | 6. 5 | 9. 6 | |
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Curriculum

Level: Kindergarten–Grade 1 Counting and cardinality
K.CC.A.1–K.CC.C.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 10 tagged checks.

Common wrong answers

1. *If they got 3: started over at one instead of counting on from the full frame*
 9. *If they got 7: counted the starting number itself as one of the extra ones*
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1. Mia’s shells: a full five-frame and 3 more.

Step 1. The frame is full, so its count is already known — there is nothing to gain by recounting it, and recounting is where children lose track.

Step 2. Start at five and touch the loose shells one at a time: “six, seven, eight”. That is $5 + 3$.

Step 3. The last number said is the answer: 8 shells.

Common wrong answer: 3 — the child restarted at one on the loose shells instead of counting on from the frame.

2. A full ten-frame and 4 loose buttons.

Step 1. A full ten-frame is 10; that is the whole point of the frame, so the count starts there.

Step 2. Count on for the loose buttons: “eleven, twelve, thirteen, fourteen”. That is $10 + 4$.

Step 3. There are 14 buttons.

Teaching note: fourteen is “ten and four more”. Hearing it that way is the first step toward place value.

3. Mia's stickers against Ben's.

Step 1. Two sets cannot be compared until each one has a single number, so finish both counts before looking at the signs.

Step 2. Mia: five, then count on “six, seven, eight” — that is $5 + 3 = 8$. Ben: four, then “five, six” — that is $4 + 2 = 6$.

Step 3. Eight is more than six, so the true sentence is Mia's stickers $\boxed{>}$ Ben's stickers.

Listen for: “Ben has more because his stickers are further apart.” Only the count decides; spacing does not.

4. How many more blocks does Kip have than Sam?

Step 1. “How many more” is a comparison, and counting on is how you compare without subtracting: start at the smaller number and walk up to the bigger one.

Step 2. Start at six: “seven, eight, nine”. Three number words were said before reaching nine.

Step 3. Kip has $\boxed{3}$ more blocks.

Listen for: counting “six, seven, eight, nine” and answering four. The starting number is where you begin, not one of the extras.

5. Jar A against Jar B.

Step 1. Both jars begin with a full ten-frame, so the tens are equal and only the loose beads can decide the comparison — but each total still has to be counted on.

Step 2. Jar A: ten, then “eleven, twelve” — that is $10 + 2 = 12$. Jar B: ten, then “eleven, twelve, thirteen, fourteen, fifteen” — that is $10 + 5 = 15$.

Step 3. Twelve is less than fifteen, so Jar A $\boxed{<}$ Jar B.

Teaching note: this is a preview of comparing teen numbers by their ones digit, because the tens already match.

6. How many more leaves does Theo have than Rosa?

Step 1. Again start at the smaller count and walk up — and this walk crosses ten, which is the part that trips children up.

Step 2. Start at eight: “nine, ten, eleven, twelve, thirteen”. Five number words were said.

Step 3. Theo has $\boxed{5}$ more leaves.

Common wrong answer: 6 — eight was counted as one of the words.

7. Ada's hearts against Leo's.

Step 1. Leo's first row is longer than Ada's, so this problem can only be settled by two finished counts, never by a glance.

Step 2. Ada: six, then count on seven more — “seven, . . . , thirteen” — so $6 + 7 = 13$. Leo: nine, then “ten, eleven, twelve” — so $9 + 3 = 12$.

Step 3. Thirteen is more than twelve, so Ada's hearts $\boxed{>}$ Leo's hearts.

Listen for: “Leo, because his row is longest.” A longer row is a guess; the count is the evidence.

8. Box 1 against Box 2.

Step 1. The two sets are grouped differently (8 and 4 against 10 and 2), so nothing can be decided until both are counted on.

Step 2. Box 1: eight, then “nine, ten, eleven, twelve” — that is $8 + 4 = 12$. Box 2: ten, then “eleven, twelve” — that is $10 + 2 = 12$.

Step 3. The two counts are the same number, so Box 1 $\equiv$ Box 2.

Teaching note: sets that look different can be equal. Being willing to write the equals sign here is real understanding, not a lucky guess.

9. How many more apples does Gus have than Ivy?

Step 1. Start at the smaller count and count on to the bigger one; the number of words said is the difference.

Step 2. Start at nine: “ten, eleven, twelve, thirteen, fourteen, fifteen”. Six number words were said.

Step 3. Gus has $\boxed{6}$ more apples.

Common wrong answer: 7 — nine was counted as one of the extras.

10. Challenge: Tray A against Tray B.

Step 1. Tray A starts from a full frame, which makes it look like the winner before anything is counted. That impression is exactly what has to be tested.

Step 2. Tray A: ten, then “eleven, twelve, thirteen” — that is $10 + 3 = 13$. Tray B: nine, then “ten, eleven, twelve, thirteen, fourteen” — that is $9 + 5 = 14$.

Step 3. Thirteen is less than fourteen, so Tray A $\boxed{<}$ Tray B.

Teaching note: the tray with the fuller frame is the smaller set. This is the strongest evidence on the page that counting on beats looking.